MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FOURTH SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
MG4DSCAQC 201 – Taxonomy and Morphology of Shellfishes

Duration: 2 Hrs. Maximum Marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer any 10 out of 12
Each question carries 2 marks

1. Define biological classification. [U][1]

2. What do you mean by phenetic classification? [U][1]

3. Mention the importance of DNA primer in barcoding analysis. [An][6]

4. Write any four external characteristics of class Crustacea. [U][2]

5. Define DNA barcoding technique in fish taxonomy. [An][6]

6. Name the major classes of phylum Mollusca. [U][3]

7. Write any four taxonomic characteristics of gastropod Mollusca. [S][4]

8. Cite the fundamental feature of phylum Mollusca. [E][4]

9. Mention the scientific name of Apple Snail. [U][5]

10. What are ink glands? Mention their uses. [S][4]

11. What is sympatric speciation? [U][1]

12. Give the common names of *Pinctada fucata* and *Crassostrea madrasensis*. [An][5]

[10 x 2=20] Marks

Part B
Short Answer Questions
Answer 8 Questions
Each question carries 5 marks

13. Differentiate the external morphology of a crustacean and a molluscan specimen. [An][2]

14. Describe the important morphological characteristics of bivalves. [U][3]

15. List the major classes under phylum Mollusca and describe features of any one class. [U][3]

16. Discuss the importance of International Code of Zoological Nomenclature (ICZN). [U][1]

17. Write an account on DNA barcoding in the classification of phylum Mollusca by specifying the amplified gene regions and primers used. [E][6]

18. Explain the basis for the classification of gastropods and describe the major taxonomic groups [U][4]

19. Describe a comparative account on the morphological characteristics of *Scylla serrata* and *Panulirus homarus*. [An][5]

20. Elaborate the economic significance of class Crustacea. [S][5]

21. Write detailed notes on Torsion in gastropods. [U][5]

22. Write an account on tautonyms, synonymy and homonymy with examples. [U][1]

[8 x 5 = 40] Marks

Part C
Essay Type Questions
Answer any One Question
Each question carries 10 marks

23. Explain the internal and external morphological characteristics of bivalves especially for their adaptation to the environment and survival. [U][3]

24. Explain the morphological features of cephalopods and explain how these structures help in locomotion, feeding and defence. [U][4]

25. Explain the species concept and discuss different kinds of species with suitable examples. [U][1]

[1 x 10 = 10] Marks